

CHAPTER 14

P.P.14.1
$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{j\omega L}{\mathbf{R} + j\omega L}$$

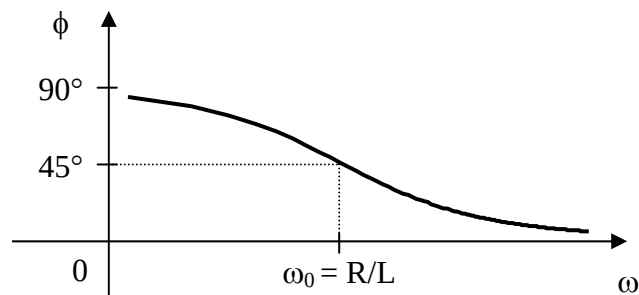
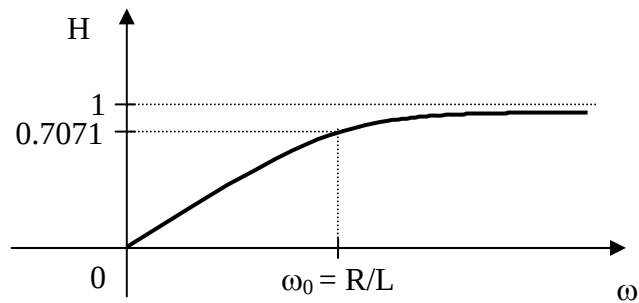
$$\mathbf{H}(\omega) = \frac{j\omega L/R}{1 + j\omega L/R} = \frac{j\omega/\omega_0}{1 + j\omega/\omega_0}$$

where $\omega_0 = \frac{R}{L}$.

$$H = |\mathbf{H}(\omega)| = \frac{\omega/\omega_0}{\sqrt{1 + (\omega/\omega_0)^2}} \qquad \phi = \angle \mathbf{H}(\omega) = \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$

At $\omega = 0$, $H = 0$, $\phi = 90^\circ$
 As $\omega \rightarrow \infty$, $H = 1$, $\phi = 0^\circ$
 At $\omega = \omega_0$, $H = \frac{1}{\sqrt{2}}$, $\phi = 90^\circ - 45^\circ = 45^\circ$

Thus, the sketches of H and ϕ are shown below.



P.P.14.2 The desired transfer function is the input impedance.

$$Z_i(s) = \frac{V_o(s)}{I_o(s)} = \left(5 + \frac{1}{s/10} \right) \parallel (3 + 2s)$$

$$Z_i(s) = \frac{(5 + 10/s)(3 + 2s)}{5 + 10/s + 3 + 2s} = \frac{5(s + 2)(s + 1.5)}{s^2 + 4s + 5}$$

The poles are at

$$p_{1,2} = \frac{-4 \pm \sqrt{16 - 20}}{2} = \underline{-2 \pm j}$$

The zeros are at

$$z_1 = \underline{-2}, \quad z_2 = \underline{-1.5}$$

P.P.14.3
$$H(\omega) = \frac{1 + j\omega/2}{(j\omega)(1 + j\omega/10)}$$

$$H_{db} = 20 \log_{10} |1 + j\omega/2| - 20 \log_{10} |j\omega| - 20 \log_{10} |1 + j\omega/10|$$

$$\phi = -90^\circ + \tan^{-1}(\omega/2) - \tan^{-1}(\omega/10)$$

The magnitude and the phase plots are shown in Fig. 14.14.

P.P.14.4
$$H(\omega) = \frac{50/400 j\omega}{(1 + j\omega/4)(1 + j\omega/10)^2}$$

$$H_{db} = -20 \log_{10} |8| + 20 \log_{10} |j\omega| - 20 \log_{10} |1 + j\omega/4| - 40 \log_{10} |1 + j\omega/10|$$

$$\phi = 90^\circ - \tan^{-1}(\omega/4) - 2 \tan^{-1}(\omega/10)$$

The magnitude and the phase plots are shown in Fig. 14.16.

P.P.14.5
$$H(\omega) = \frac{10/400}{(j\omega) \left(1 + \frac{j\omega 8}{40} + \left(\frac{j\omega}{20} \right)^2 \right)}$$

$$H_{db} = -20 \log_{10} |40| - 20 \log_{10} |j\omega| - 20 \log_{10} |1 + j\omega/5 - \omega^2/400|$$

$$\phi = -90^\circ - \tan^{-1} \left(\frac{0.2\omega}{1 - \omega^2/400} \right)$$

The magnitude and the phase plots are shown in Fig. 14.18.

P.P.14.6

A zero at $\omega = 0.5$, $1 + j\omega/0.5$

A pole at $\omega = 1$, $\frac{1}{1 + j\omega/1}$

Two poles at $\omega = 10$, $\frac{1}{(1 + j\omega/10)^2}$

Hence,

$$H(\omega) = \frac{1 + j\omega/0.5}{(1 + j\omega/1)(1 + j\omega/10)^2} = \frac{(1/0.5)(0.5 + j\omega)}{(1/100)(1 + j\omega)(10 + j\omega)^2}$$

$$H(\omega) = \frac{200(s + 0.5)}{(s + 1)(s + 10)^2}$$

P.P.14.7

$$(a) \quad Q = \frac{\omega_0 L}{R} \longrightarrow \omega_0 = \frac{QR}{L} = \frac{(50)(4)}{25 \times 10^{-3}} = 8 \times 10^3 \text{ rad/s}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \longrightarrow C = \frac{1}{\omega_0^2 L} = \frac{1}{(64 \times 10^6)(25 \times 10^{-3})}$$

$$C = \underline{\underline{0.625 \mu\text{F}}}$$

$$(b) \quad B = \frac{\omega_0}{Q} = \frac{8 \times 10^3}{50} = \underline{\underline{160 \text{ rad/s}}}$$

Since $Q > 10$,

$$\omega_1 = \omega_0 - \frac{B}{2} = 8000 - 80 = \underline{\underline{7920 \text{ rad/s}}}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 8000 + 80 = \underline{\underline{8080 \text{ rad/s}}}$$

$$(c) \quad \text{At } \omega = \omega_0, \quad P = \frac{V_{in}^2}{2R} = \frac{100^2}{8} = \underline{\underline{1.25 \text{ kW}}}$$

$$\text{At } \omega = \omega_1, \quad P = 0.5 \cdot \frac{V_{in}^2}{2R} = \underline{\underline{0.625 \text{ kW}}}$$

$$\text{At } \omega = \omega_2, \quad P = 0.5 \cdot \frac{V_{in}^2}{2R} = \underline{\underline{0.625 \text{ kW}}}$$

$$\text{P.P.14.8} \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(20 \times 10^{-3})(5 \times 10^{-9})}} = 10^5 = \underline{\underline{100 \text{ krad/s}}}$$

$$Q = \frac{R}{\omega_0 L} = \frac{100 \times 10^3}{(10^5)(20 \times 10^{-3})} = \underline{\underline{50}}$$

$$B = \frac{\omega_0}{Q} = \frac{10^5}{50} = \underline{\underline{2 \text{ krad/s}}}$$

Since $Q > 10$,

$$\omega_1 = \omega_0 - \frac{B}{2} = 100,000 - 1,000 = \underline{\underline{99 \text{ krad/s}}}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 100,000 + 1,000 = \underline{\underline{101 \text{ krad/s}}}$$

$$\text{P.P.14.9} \quad Z = j\omega 1 + 10 \parallel \frac{1}{j\omega 0.2} = j\omega + \frac{50}{5 + j10\omega}$$

$$Z = j\omega + \frac{10(1 - j2\omega)}{1 + 4\omega^2}$$

$$\text{Im}(Z) = 0 \longrightarrow \omega - \frac{20\omega}{1 + 4\omega^2} = 0$$

$$\omega = \frac{20\omega}{1 + 4\omega^2} \longrightarrow 1 + 4\omega^2 = 20$$

$$\omega = \frac{\sqrt{19}}{2} = \underline{\underline{2.179 \text{ rad/s}}}$$

$$\text{P.P.14.10} \quad H(s) = \frac{V_o}{V_i} = \frac{R_2 \parallel sL}{R_1 + R_2 \parallel sL}, \quad s = j\omega$$

$$H(s) = \frac{sR_2 L}{R_1 R_2 + sR_1 L + sR_2 L}$$

$$H(\omega) = \frac{j\omega R_2 L}{R_1 R_2 + j\omega L (R_1 + R_2)}$$

$$H(0) = 0$$

$$H(\omega) = \lim_{\omega \rightarrow \infty} \frac{jR_2 L}{R_1 R_2 / \omega + jL (R_1 + R_2)} = \frac{R_2}{R_1 + R_2}$$

i.e. a **highpass filter**.

The corner frequency occurs when $H(\omega_c) = \frac{1}{\sqrt{2}} \cdot H(\infty)$.

$$\mathbf{H}(\omega) = \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{j\omega L}{j\omega L + R_1 R_2 / (R_1 + R_2)} \right)$$

$$\mathbf{H}(\omega) = \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{j\omega}{j\omega + k} \right), \quad \text{where } k = \frac{R_1 R_2}{(R_1 + R_2)L}$$

At the corner frequency,

$$\frac{1}{\sqrt{2}} \cdot \frac{R_2}{R_1 + R_2} = \frac{R_2}{R_1 + R_2} \cdot \left| \frac{j\omega_c}{j\omega_c + k} \right|$$

$$\frac{1}{\sqrt{2}} = \frac{\omega_c}{\sqrt{\omega_c^2 + k^2}} \longrightarrow \omega_c = k = \frac{R_1 R_2}{(R_1 + R_2)L}$$

Hence,
$$\mathbf{H}(\omega) = \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{j\omega}{j\omega + \omega_c} \right)$$

and the corner frequency is

$$\omega_c = \frac{(100)(100)}{(100 + 100)(2 \times 10^{-3})} = \underline{\underline{25 \text{ krad/s}}}$$

P.P.14.11 $B = 2\pi(20.3 - 20.1) \times 10^3 = 400\pi$

Assuming high Q,

$$\omega_0 = \frac{\omega_1 + \omega_2}{2} = \frac{(2\pi)(40.4 \times 10^3)}{2} = 40.4\pi \times 10^3 \text{ rad/s}$$

$$Q = \frac{\omega_0}{B} = \frac{40.4\pi \times 10^3}{400\pi} = \underline{\underline{101}}$$

$$B = \frac{R}{L} \longrightarrow L = \frac{R}{B} = \frac{20 \times 10^3}{400\pi} = \underline{\underline{15.916 \text{ H}}}$$

$$Q = \frac{1}{\omega_0 CR} \longrightarrow C = \frac{1}{\omega_0 QR}$$

$$C = \frac{1}{(40.4\pi \times 10^3)(101)(20 \times 10^3)} = \underline{\underline{3.9 \text{ pF}}}$$

P.P.14.12 Given $H(\infty) = 5$ and $f_c = 2 \text{ kHz}$

$$\omega_c = 2\pi f_c = \frac{1}{R_i C_i}$$

$$R_i = \frac{1}{2\pi f_c C_i} = \frac{1}{(2\pi)(2 \times 10^3)(0.1 \times 10^{-3})}$$

$$R_i = 795.8 \cong \underline{\underline{800 \Omega}}$$

$$H(\infty) = \frac{-R_f}{R_i} = -5 \longrightarrow R_f = 5R_i = 3,978 \cong \underline{\underline{4 \text{ k}\Omega}}$$

P.P.14.13 $Q = 10$, $\omega_0 = 20 \text{ krad/s}$

$$B = \frac{\omega_0}{Q} = 2 \text{ krad/s}$$

$$\omega_1 = \omega_0 - \frac{B}{2} = 19 \text{ krad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 21 \text{ krad/s}$$

Since $\omega_1 = \frac{1}{C_2 R}$,

$$C_2 = \frac{1}{\omega_1 R} = \frac{1}{(19 \times 10^3)(10 \times 10^3)} = \underline{\underline{5.263 \text{ nF}}}$$

$$C_1 = \frac{1}{\omega_2 R} = \frac{1}{(21 \times 10^3)(10 \times 10^3)} = \underline{\underline{4.762 \text{ nF}}}$$

$$K = \frac{R_f}{R_i} = 5 \longrightarrow R_f = 5R_i = \underline{\underline{50 \text{ k}\Omega}}$$

P.P.14.14 $K_f = \frac{\omega'_c}{\omega_c} = \frac{2\pi \times 10^4}{1} = 2\pi \times 10^4$

$$C' = \frac{C}{K_m K_f} \longrightarrow K_m = \frac{C}{C' K_f} = \frac{1}{(15 \times 10^{-9})(2\pi \times 10^4)} = \frac{10^4}{3\pi}$$

$$R' = K_m R = \frac{10^4}{3\pi} (1) = 1.061 \text{ k}\Omega$$

$$L' = \frac{K_m}{K_f} L = \frac{10^4}{3\pi} \cdot \frac{2}{2\pi \times 10^4} = 33.77 \text{ mH}$$

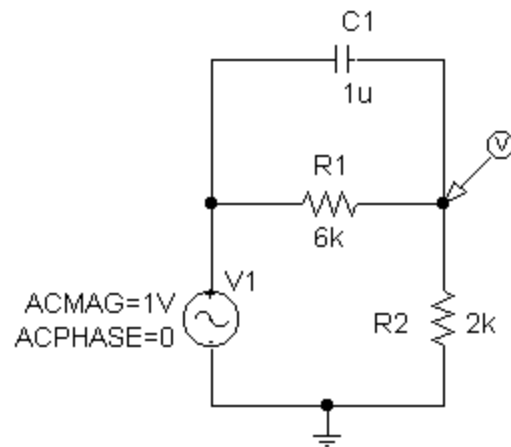
Therefore,

$$R'_1 = R'_2 = \underline{\underline{1.061 \text{ k}\Omega}}$$

$$C'_1 = C'_2 = \underline{\underline{15 \text{ nF}}}$$

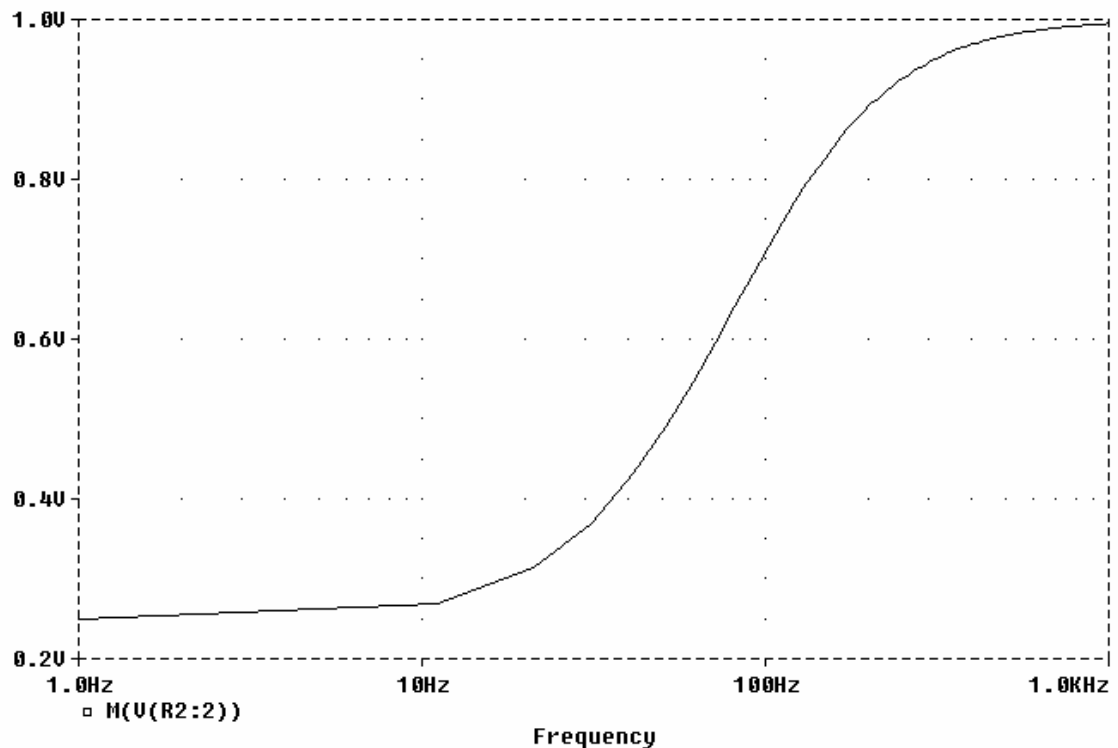
$$L' = \underline{\underline{33.77 \text{ mH}}}$$

P.P.14.15 The schematic is shown in Fig. (a).

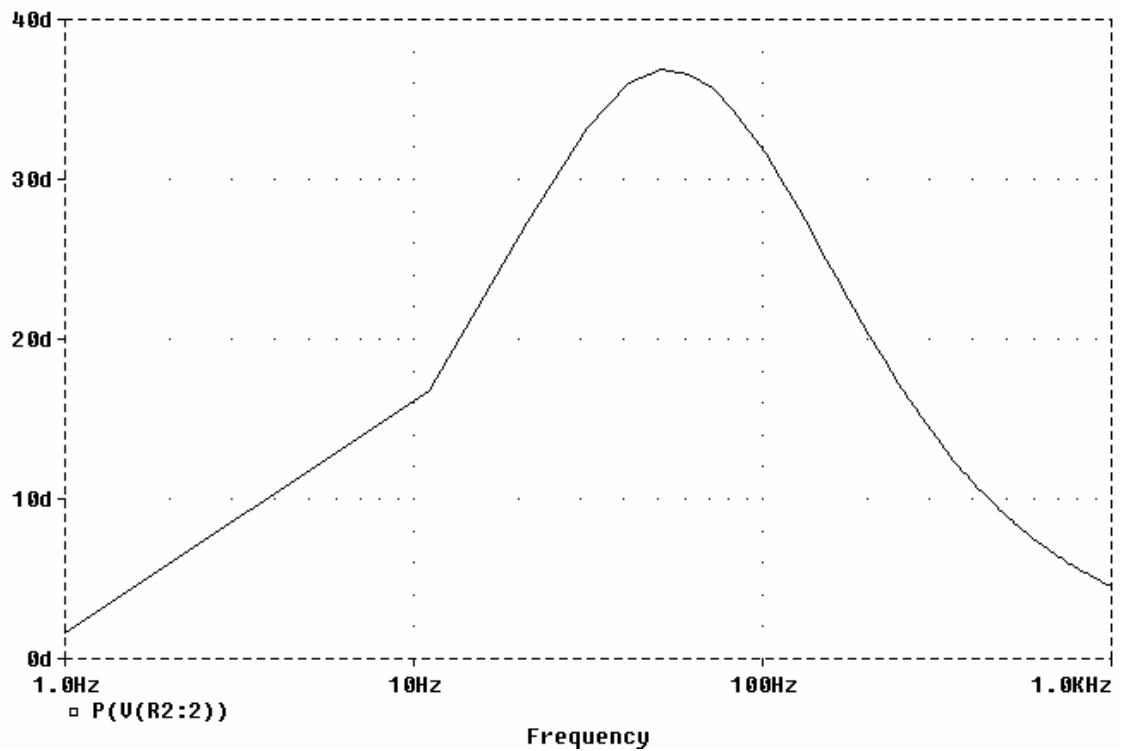


(a)

Use the **AC Sweep** option of the **Analysis Setup**. Choose a *Linear* sweep type with the following *Sweep Parameters* : *Total Pts* = 100, *Start Freq* = 1, and *End Freq* = 1K. After saving and simulating the circuit, we obtain the magnitude and phase plots are shown in Figs. (b) and (c).

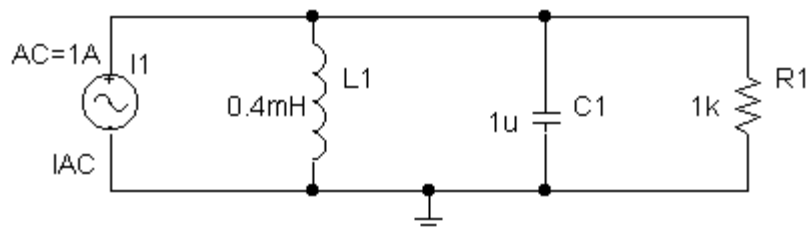


(b)



(c)

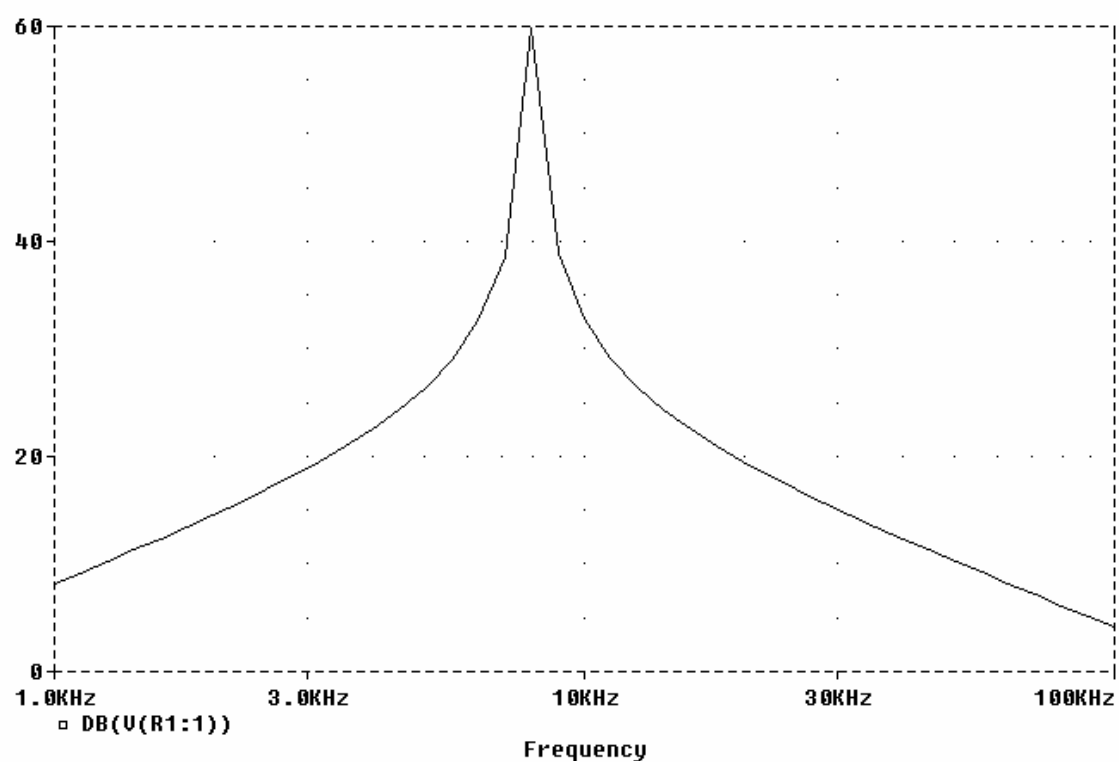
P.P.14.16 The schematic is shown in Fig. (a).



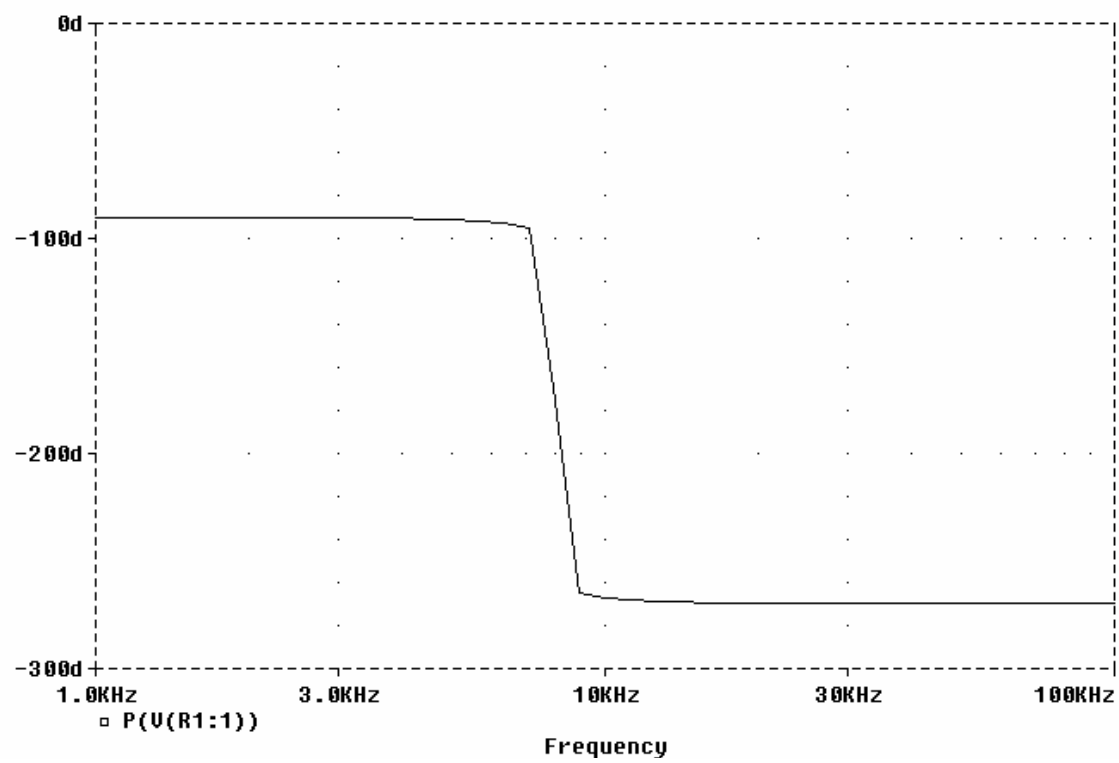
(a)

Use the **AC Sweep** option of the **Analysis Setup**. Choose a *Decade* sweep type with these *Sweep Parameters* : *Pts/Decade* = 20, *Start Freq* = 1K, and *End Freq* = 100K. Save and simulate the circuit.

For the magnitude plot, choose DB() from the **Analog Operators and Functions** list. Then, select the voltage V(R1:1) and OK. Another option would be to type DB(V(R1:1)) as the **Trace Expression**. For the phase plot, choose P() from the **Analog Operators and Functions** list. Then, select the voltage V(R1:1) and OK. Another option would be to type VP(R1:1) as the **Trace Expression**. The resulting magnitude and phase plots are shown in Figs. (b) and (c).



(b)



(c)

P.P.14.17 $\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$

or $C = \frac{1}{4\pi^2 f_0^2 L}$

For the high end of the band, $f_0 = 108 \text{ MHz}$

$$C_1 = \frac{1}{4\pi^2 (108^2 \times 10^{12})(4 \times 10^{-6})} = 0.543 \text{ pF}$$

For the low end of the band, $f_0 = 88 \text{ MHz}$

$$C_2 = \frac{1}{4\pi^2 (88^2 \times 10^{12})(4 \times 10^{-6})} = 0.818 \text{ pF}$$

Therefore, C must be adjustable and be in the range **0.543 pF to 0.818 pF** .

P.P.14.18

For BP_6 , $f_0 = 1336 \text{ Hz}$ and it passes frequencies in the range $1209 \text{ Hz} < f < 1477 \text{ Hz}$.

$$B = 2\pi(1477 - 1209) = 1683.9$$

$$L = \frac{R}{B} = \frac{600}{1683.9} = \underline{\underline{0.356 \text{ H}}}$$

$$C = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2 (1336)^2 (0.356)} = \underline{\underline{39.83 \text{ nF}}}$$

P.P.14.19 $C = 10 \text{ } \mu\text{F}$ and $R_1 = R_2 = 8 \text{ } \Omega$

$$2\pi f_c = \frac{1}{R_1 C} \longrightarrow f_c = \frac{1}{2\pi R_1 C} = \frac{1}{(2\pi)(8)(10 \times 10^{-6})} = \underline{\underline{1.989 \text{ kHz}}}$$

$$L = \frac{R_2}{2\pi f_c} = \frac{8}{(2\pi)(1.989 \times 10^3)} = \underline{\underline{0.64 \text{ mH}}}$$